

ASEN 5090 Lecture 9

Tropospheric Effects

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GNSS PSEUDORANGE CORRECTIONS

Assigned: 23 September 2026

Due: 5 October 2026 (Courtesy late submission 7 October)

OBJECTIVES

In this assignment we compute and apply satellite clock, relativity, ionospheric delay, and tropospheric delay corrections to the range calculations completed in HW3 and compare multipath errors for different pseudorange types.

We will work with data from PRN14 on August 19, 2026. PRN14 is assigned to SVN77 – a Block III GPS satellite that transmits C/A, P(Y), L1C, L2 and L5 civil signals.

OVERVIEW

- Date: Wednesday, August 19, 2026 = GPS Week 2432 (or 384), TOW 259200-345600/ DOY 231
- User location: NIST (in Boulder, CO): ECEF coordinates: [-1288398.567 -4721696.932 4078625.350] (m)]
- RINEX File: NIST00USA_R_20262310000_01D_30S_MO.rnx
- Broadcast ephemeris file: brdcDOY0.26n (DOY is the 3-digit day of year)
- Hints – I usually read in the rinex file once and then save the data into a .mat file which I then load to speed things up for subsequent runs. Also, when you are first getting started it can be easier to work with a small subset of the data instead of the entire day.

ASSIGNMENT

1. **Starting point** (10pts): Select all the data for **PRN14** from the RINEX obs file to work with for this assignment. Create a time vector just including the times for which you have an observation for the satellite. Compute the expected range (R) for each of these times using your code from HW3.

Compute and plot the difference between the measured C/A code pseudorange (C1C) and the expected range from HW3 (R in meters) versus time for our reference date. *What should it look like?*

$$dPRO = C1C - R$$

Print out the first and last values of dPRO.

2. **Clock error** (10pts): Plot the satellite clock correction (bsv) in meters from the `eph2pvt` function for each measurement time. *What should it look like?*

$$\text{Compute and plot: } dPR1 = C1C - (R - \text{bsv})$$

Print out the first and last values of dPR1.

3. **Relativity** (10pts): Add code to the `eph2pvt` function to compute the relativistic correction (relsv) (meters) based on the orbital elements or GPS position and velocity based on the ephemeris data. Use the negative sign as given in Lecture to be compatible with the IS-GPS documentation.

Compute and plot the relativistic correction (meters) for each measurement time. *What should it look like?*

$$\text{Compute and plot: } dPR2 = C1C - (R - \text{bsv} - \text{relsv})$$

Print out the first and last values of dPR2.

4. **Troposphere** (15pts): Write a function to compute the simple tropospheric delay model (meters) based on the satellite elevation and an assumed zenith delay for the ground station location. For the NIST ground station you may assume the zenith value $z_d = 2$ m. Your function call should look like:

```
[tropo] = tropomodel(zd, elevation)
```

Compute and plot the tropo correction (meters) for each measurement time. *What should it look like?*

Compute and plot $dPR3 = C1C - (R - bsv - relsv + tropo)$

Print out the first and last values of dPR3.

Optional (up to 3pts): Download and implement the UNB3 model

<https://gge.ext.unb.ca/Resources/unb3m/unb3m.html>

Compute and plot the UNB3 tropo correction on the same graph as the simple model.

Compute and plot $dPR3UNB = C1C - (R - bsv - relsv + tropoUNB)$

Print out the first and last values of dPR3UNB.

5. **Ionosphere** (15pts): For the ionospheric correction, rather than using a model-based correction (like Klobuchar) we will combine GPS pseudorange measurements into an ionosphere free combination.

Write a function to compute the iono-free combination of the pseudoranges reported in the RINEX obs file and also the iono correction based on these two measurements. *To allow your function to be used for any combination of measurements pass into it the carrier frequencies ($f1 = 1575.42e6$, $f2 = ?$, $f5 = ?$) as well as the measurements.* Your function call to work with C1C and C2L should look like:

```
[PRIF12, iono] = ionocorr (C1C, f1, C2L, f2)
```

Or to combine C1C and C5Q signals

```
[PRIF15, iono] = ionocorr (C1C, f1, C5Q, f5)
```

Compute and plot the iono correction (meters) based on **C1C** and **C2L** for each measurement time. *What should it look like?*

Compute and plot $dPR4 = PRIF - (R - bsv - relsv + tropo)$

* Notice here we are changing the measurement, not the model like we did in the previous parts.

Print out the first and last values of dPR4.

Optional (up to 3pts): Compute an alternative dual frequency combination using a combination of carrier phase data on two frequencies (CPIF)

Compute and plot the iono correction for your new combo.

Compute and plot $dCP4alt = CPIF - (R - bsv - relsv + tropo)$

Print out the first and last values of dCP4alt.

6. **Discussion** (20pts): **Once everything is working, plot dPR1, dPR2, dPR3, and dPR4 vs time on the same graph**
Discuss each step of your results, commenting on the magnitude and other characteristics of each error source you corrected. Note any unexpected features of your plots. If the results don't turn out the way you expect, describe what you did to investigate possible problems in your code or in the data.

Optional (up to 2 pts): Discuss the impact of UNB3 and/or carrier phase iono corrections.

7. **Multipath** (20pts): To investigate pseudorange multipath on different types of pseudoranges available for the newest GPS satellites, write a function to compute the multipath observable and code-minus-carrier (Lect15 slide31) for a given combination of code & carrier measurements. The function call should look like this:

```
[MP1, CMC1] = mpath(C1C, L1C, f1, L2W, f2)
```

Compute and plot the CMC results for these four codes: C1C, C2W, C2L, C5Q for all the measurement times.

On a separate subplot, show the corresponding signal to noise measurements (S1C, S2W, S2L, S5Q).

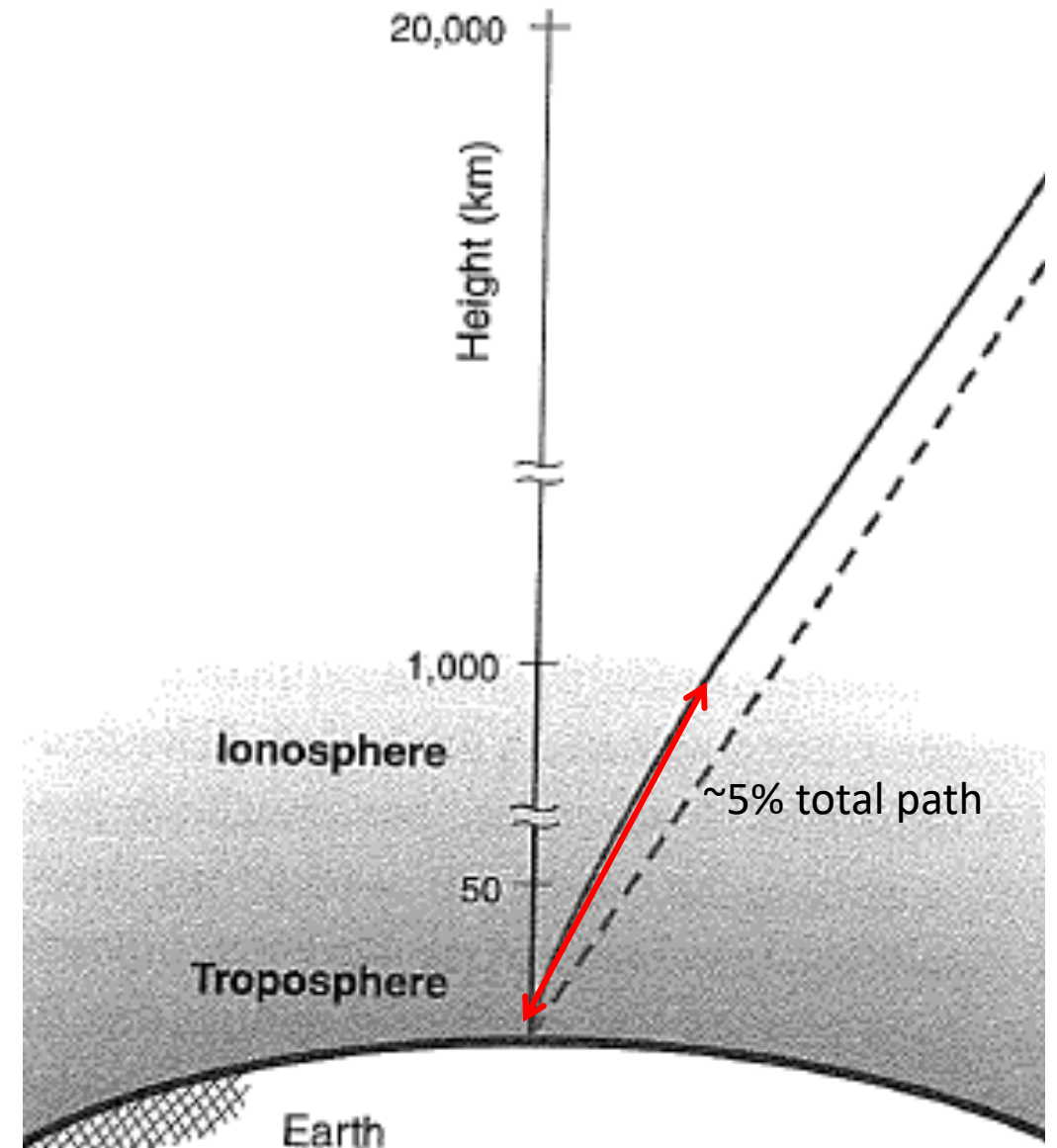
For each code, choose which second signal to use for the iono corrections and plot the MP results.

Briefly describe the results. Which measurements would you choose to use for positioning?

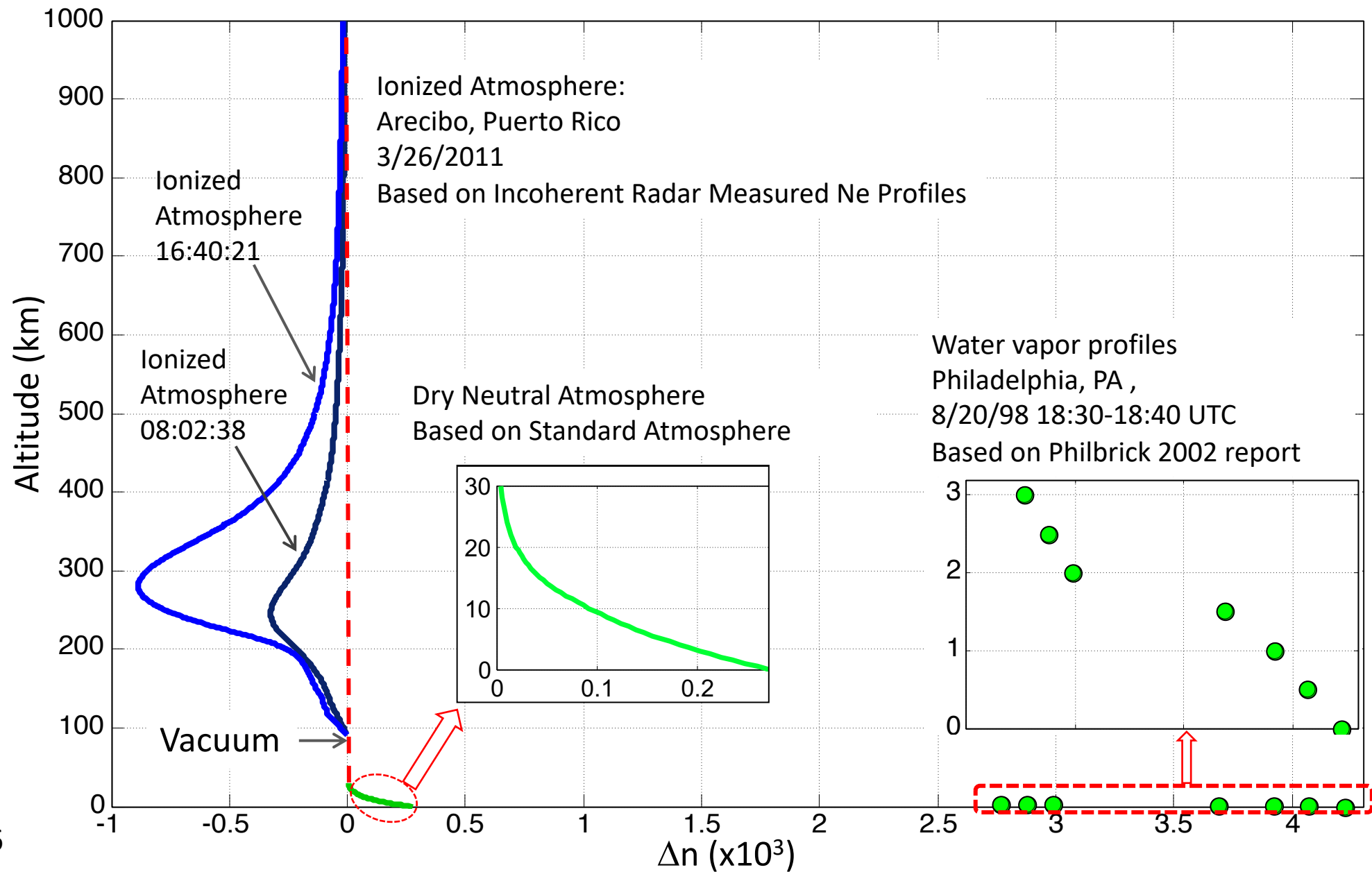
GPS Signal Propagation

- GPS signals are refracted in the Earth's atmosphere
- Bending is small for most ray paths
- Change in speed of propagation affects pseudorange and phase measurements
- Magnitude of effect is (somewhat) proportional to the propagation distance, in other words, has a dependence on the incidence angle.
- Net effect as a delay:

$$\Delta\tau = \tau - \tau_{vacuum} = \frac{1}{c} \int_s^R (n(l) - 1) dl$$



Neutral & Ionized Atmosphere Refractive Index



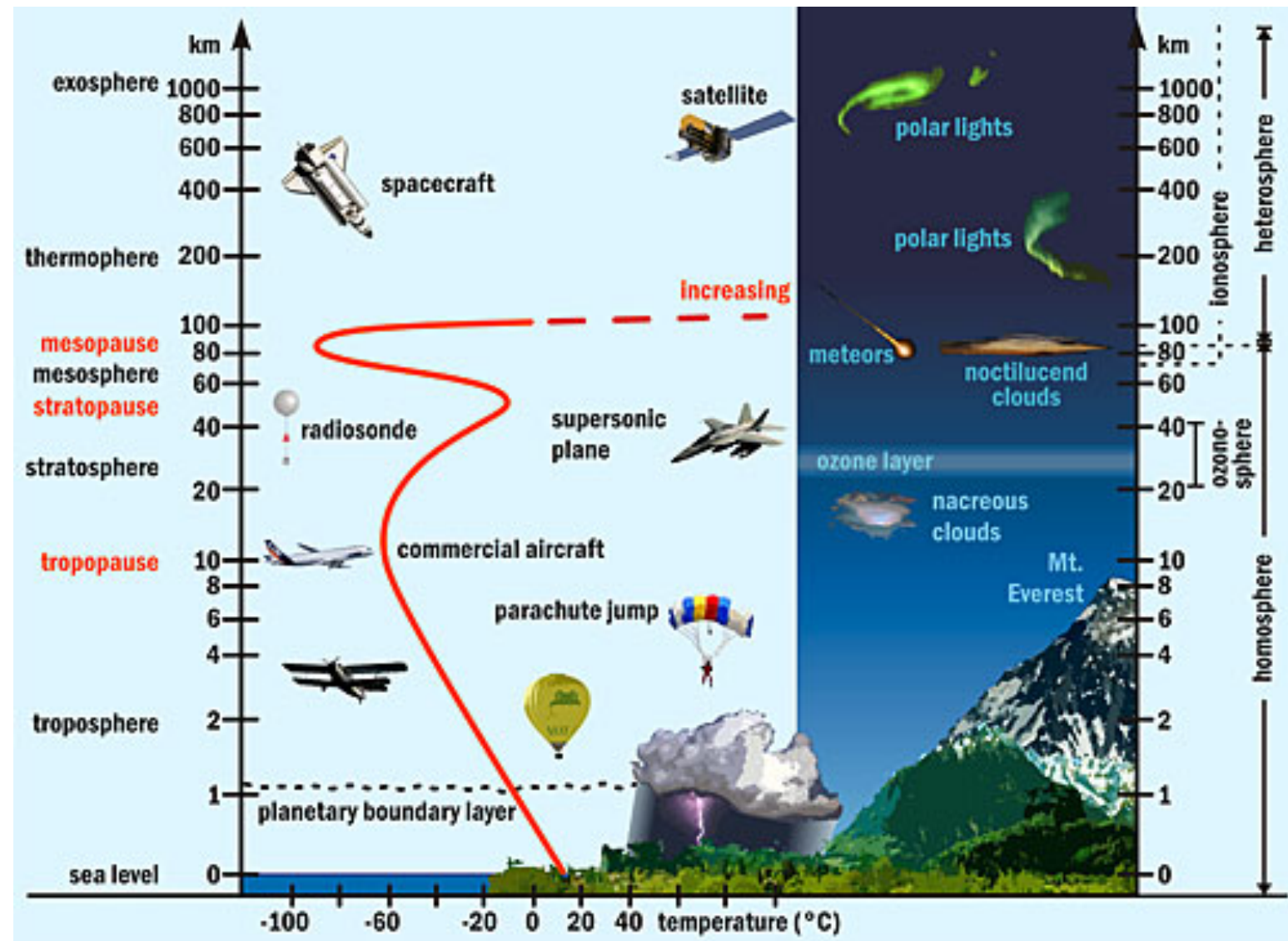
From PNT Ch 31, Fig 31.6

Outline

- Introduction to troposphere
- Hydrostatic (dry) delay and water vapor (wet) delay
- Mapping functions
- Tropospheric delay modeling and correction
- GNSS contributions to tropospheric mapping



Pictorial representation of atmospheric structures

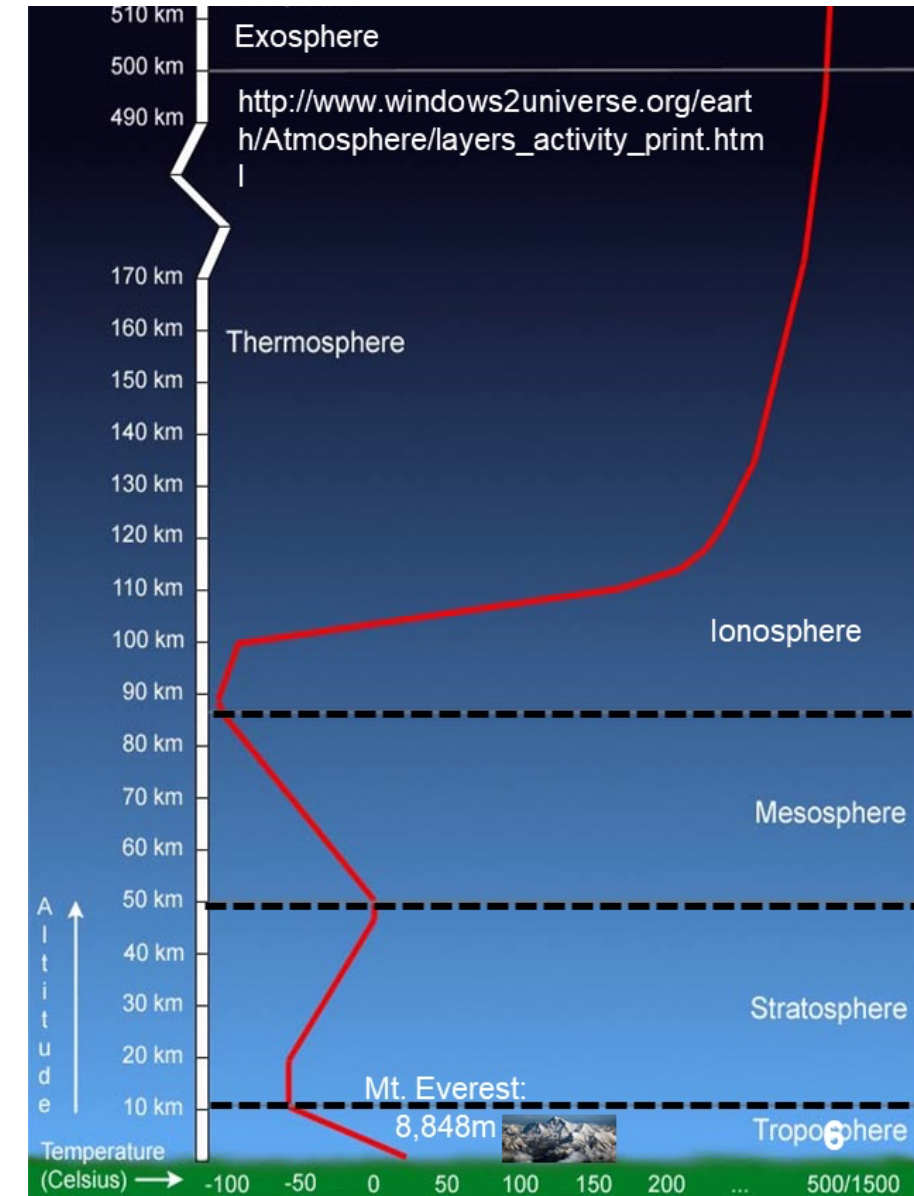
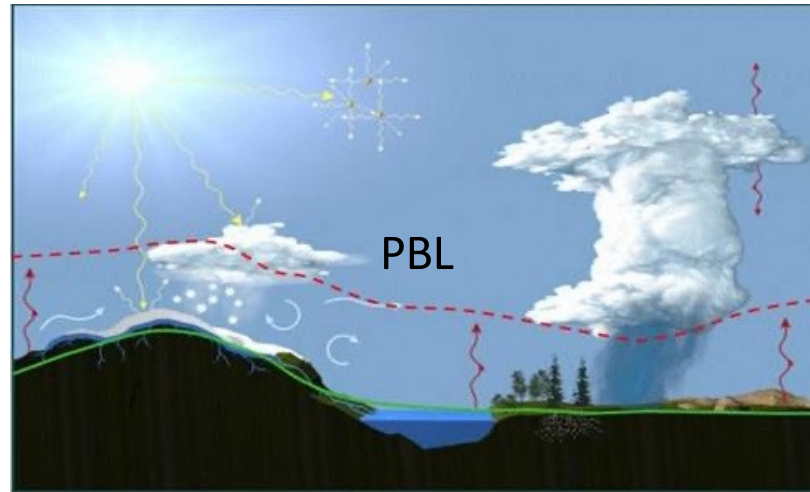


Francisco Colomer DOI: <https://doi.org/10.22323/1.065.0003>

Figure 1: Structure of the Earth's atmosphere.

Troposphere Properties

- Generates all the weather and clouds
- Shallow cover of entire Earth
 - 7~17 km thick (deep at the tropics, shallow at the poles)
 - 75% atmosphere mass
 - 99% atmosphere water vapor and aerosols
- Planetary boundary layer (PBL)
 - Separates “turbulent atmosphere” due to contact with planetary surface from “free atmosphere” above
 - ~200m - 3km altitude
 - Depends on location/time



Refractive Index and Range Error

“Slowing down” the radio wave

$$n = \frac{c}{v}$$

c ← Speed of light in vacuum
 v ← Speed of light in the medium

$$v = \frac{dl}{dt} = \frac{c}{n(l)} \quad \longrightarrow \quad dt = \frac{n(l)}{c} dl$$

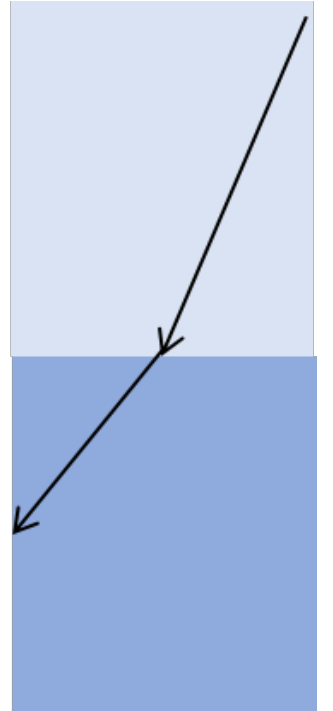
$$\tau = \int_S^R \frac{n(l)}{c} dl = \frac{1}{c} \int_S^R n(l) dl$$

$$\Delta\tau = \tau - \tau_{vacuum} = \frac{1}{c} \int_S^R (n(l) - 1) dl$$

range error $\Delta\rho$

Bending: very small except at low elevations

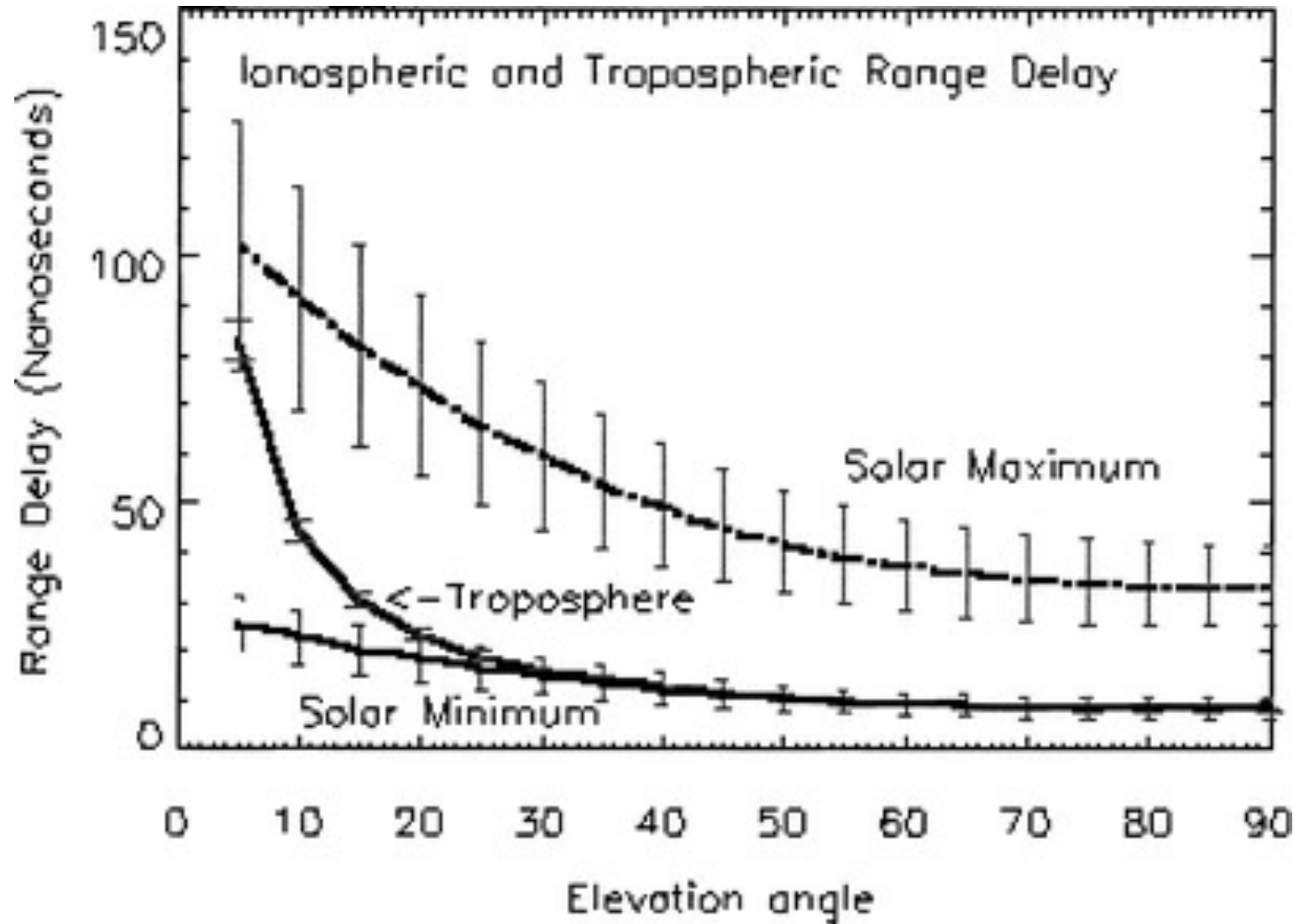
Usually not considered except in ray tracing, radio occultation, and some tropospheric delay models for elev. $< \sim 5$ degrees.



Basic Error Characteristics

- Dry gases (N_2 , O_2) and water vapor cause GPS signal refraction
 - Water vapor: below 12 km altitude. Varies with local weather. Difficult to model.
 - Dry gases: below 16 km near equator, 9 km near pole. Contribute to 90% troposphere refraction error. Can be modeled.
- Is not frequency dependent
- Has the same effect on code and carrier
- Error ranges from about 2 to 25m (zenith to low-elevation)
- Less variable than ionosphere error

Comparison of Mid-latitude Troposphere and Ionosphere Error



Klobuchar and Kunches,
"Comparative range delay and
variability in the earth's
troposphere and the ionosphere,"
GPS Solutions, 7, 55-58, 2003.

Dry and Wet Tropospheric Effects

- Refractive index n & Refractivity N :

$$N = (n - 1) \times 10^6$$

- Dry and wet delays are modeled separately:

$$N = N_d + N_w$$

$$\tilde{T} = 10^{-6} \int_{h_t}^0 N dl = 10^{-6} \int_{h_t}^0 (N_d + N_w) dl = \tilde{T}_d + \tilde{T}_w$$

$$N_d = 77.64 \frac{P}{T} \quad N_w = 3.73 \times 10^5 \frac{e}{T^2}$$

P : atmospheric pressure (mB)

e : partial pressure of water vapor (mB)

T : temperature (K)

Tropospheric Delay Models

Zenith delay: $\tilde{T}_z = \tilde{T}_{z,d} + \tilde{T}_{z,w}$

1. Saastamoinen model
2. Hopfield model

Line-of-sight (LOS) delay with mapping functions:

$$\tilde{T} = \tilde{T}_{z,d} m_d(el) + \tilde{T}_{z,w} m_w(el)$$

Obliquity factors
(one common factor for simple models)



Saastomoinen Model

$$\tilde{T}_{z,d} = 0.002277(1 + 0.0026 \cos(2\varphi) + 0.00028H)P_0 \quad (\text{mm})$$

$$\tilde{T}_{z,w} = 0.002277 \left(\frac{1255}{T_0} + 0.05 \right) e_0 \quad (\text{mm})$$

Values at RX antenna
based on local measurements
or Standard Atmosphere model

φ : latitude
 H : Antenna height above sea level (km) (orthometric height)

$\left\{ \begin{array}{l} P_0: \text{total pressure (mB)} \\ e_0: \text{partial water vapor pressure (mB)} \\ T_0: \text{temperature (K)} \end{array} \right.$

$$e_0 = 6.108 \times RH \times e^{\frac{17.15T_0 - 4684}{T_0 - 38.45}}$$

↑ Relative humidity (it is a percentage value)



Hopfield Model

$$N_d(h) = N_{d0} \left(1 - \frac{h}{h_d}\right)^4$$

$$N_w(h) = N_{w0} \left(1 - \frac{h}{h_w}\right)^4$$

$$h_d \approx 43km$$

$$h_w \approx 12km$$

N_{d0} : dry refractivity at the surface, $N_d(h_d) = 0$

N_{w0} : wet refractivity at the surface, $N_w(h_w) = 0$

$$\tilde{T}_{z,d} = 77.6 \times 10^{-6} \frac{P_0}{T_0} \frac{h_d}{5}$$

$$\tilde{T}_{z,w} = 0.373 \frac{e_0}{T_0^2} \frac{h_w}{5}$$

$$\tilde{T}_z = \tilde{T}_{z,d} + \tilde{T}_{z,w}$$

P_0 : total pressure (mB)

e_0 : partial water vapor pressure (mB)

T_0 : temperature (K)



Example Numbers

$\tilde{T}_{z,d}$ can be modeled with accuracy at **mm** level, given accurate surface pressure measurements

$\tilde{T}_{z,w}$ is more variable, harder to model.
Typical accuracy is at **1-2 cm** level, using local weather instrument measurements.
Model-based prediction, accuracy usually **5-10 cm**.

$\tilde{T}_{z,d}$ typical values:
2.3~2.6 m at sea level
2 m at Denver
1 m at the peak of Himalayas

$\tilde{T}_{z,w}$ typical values: **0-80 cm**
mm in polar regions, **cm** in desert, **10's** cm in tropical areas

IGS Tropospheric Delay

- IGS estimates tropospheric zenith delay for each station D
- ZPD files: estimates of zenith delay for each station 5 minute intervals

KOKV

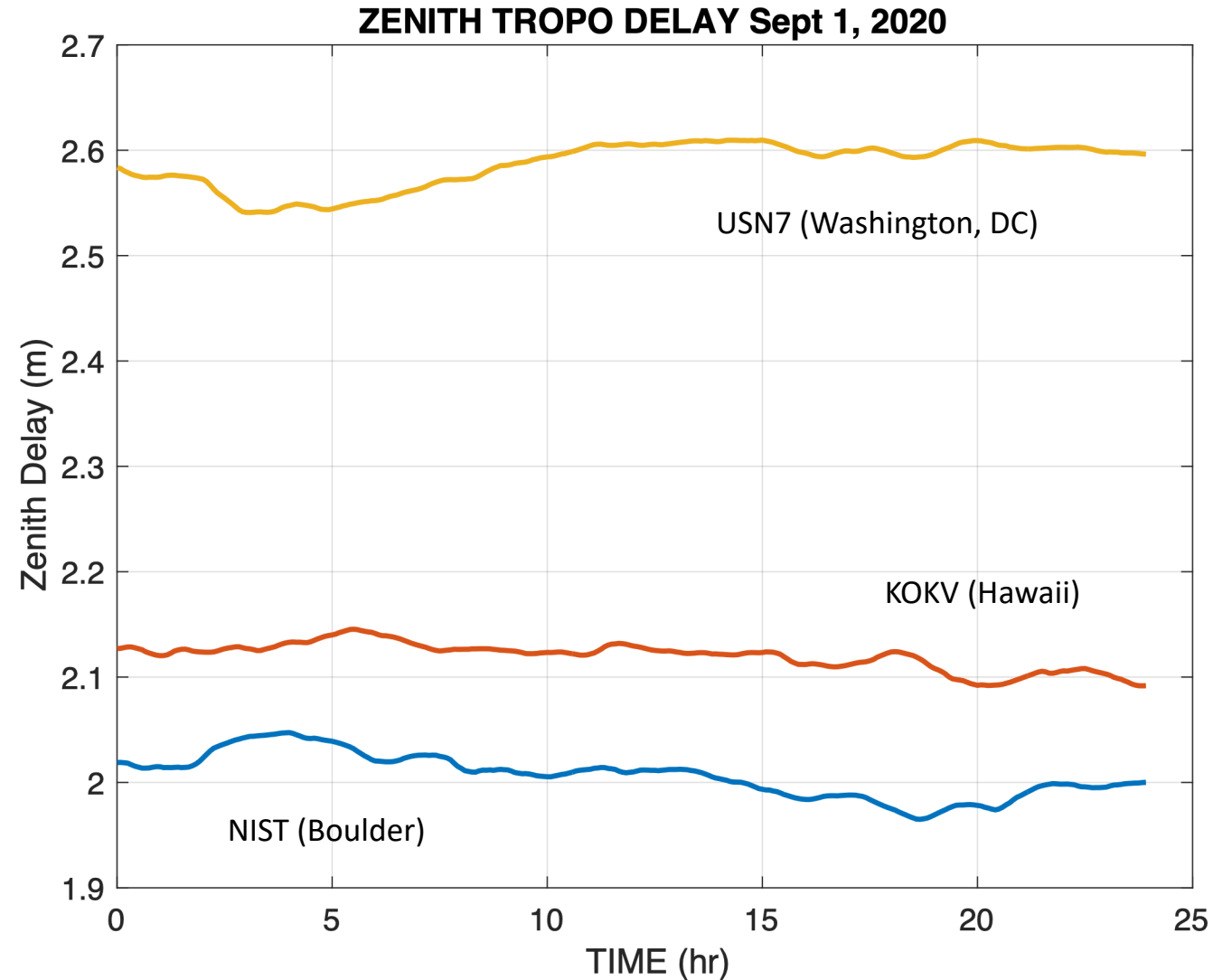
Latitude (N is +) : +22 07 34.5496

Longitude (E is +) : -159 39 53.7472

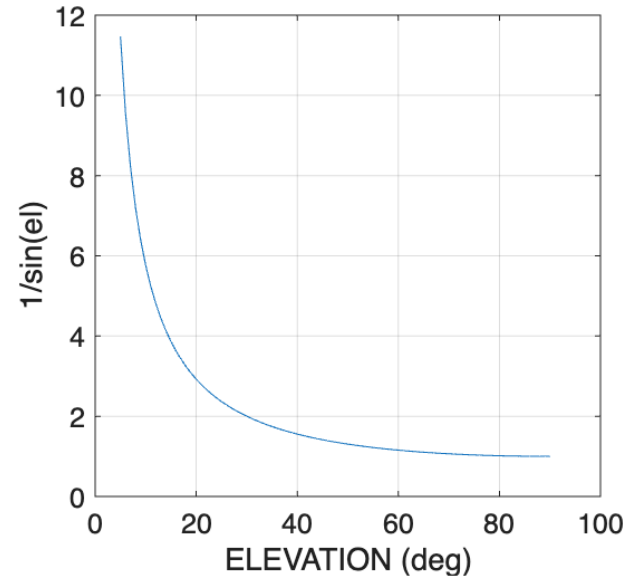
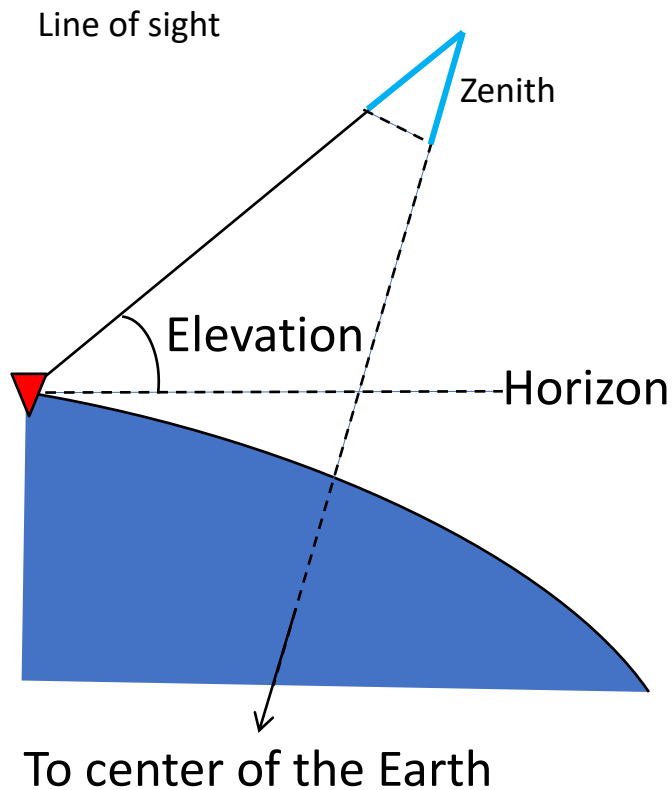
Elevation (m,ellips.) : **1167.5216 (high altitude)**

Kokee Park, Waimea, Kauai. Kauai

<https://cddis.nasa.gov/archive/gnss/products/troposphere/zpd/>



Mapping Function: Obliquity Factor



$$m(el) = \frac{1}{\sin(el)} \quad \text{Flat surface approximation}$$

$$m_d(el) = \frac{1}{\sin(el) + \frac{0.00143}{\tan(el) + 0.0445}}$$

$$m_w(el) = \frac{1}{\sin(el) + \frac{0.00035}{\tan(el) + 0.017}}$$

$$i = d, w \quad m_i(el) = \frac{1 + \frac{a_i}{1 + \frac{b_i}{1 + c_i}}}{\sin(el) + \frac{a_i}{\sin(el) + \frac{b_i}{\sin(el) + c_i}}}$$

a, b, c : coefficients empirically determined

(T. A. HERRIN, 1992)

Mapping Function: compare tropo & iono

TROPO

$$m(el) = \frac{1}{\sin(el)} \quad \text{Flat surface approximation}$$

$$m(el) = \frac{1}{\sqrt{(1 - (\cos(el)/1.001))^2}} \quad (\text{ME 5.42})$$

Separate mapping for dry and wet components

$$m_d(el) = \frac{1}{\sin(el) + \frac{0.00143}{\tan(el) + 0.0445}}$$

$$m_w(el) = \frac{1}{\sin(el) + \frac{0.00035}{\tan(el) + 0.017}}$$

IONO

$$OF_I = \frac{1}{\cos \zeta'} = \frac{1}{\sqrt{1 - \left(\frac{R_E \sin \zeta}{R_E + h_I}\right)^2}}$$

$$OF_I \approx 1.0 + 16.0 \times (0.53 - el)^3$$

Careful with this approximation – el is in semi-circles



UNB3 (or UNB3m)

<https://gge.ext.unb.ca/Resources/unb3m/unb3m.html>

Table 1. Look-up table of UNB3 model.

Average					
Latitude (degrees)	Pressure (mbar)	Temperature (K)	WVP* (mbar)	β (K m ⁻¹)	λ (-)
15	1013.25	299.65	26.31	6.30e-3	2.77
30	1017.25	294.15	21.79	6.05e-3	3.15
45	1015.75	283.15	11.66	5.58e-3	2.57
60	1011.75	272.15	6.78	5.39e-3	1.81
75	1013.00	263.65	4.11	4.53e-3	1.55
Amplitude					
Latitude (degrees)	Pressure (mbar)	Temperature (K)	WVP* (mbar)	β (K m ⁻¹)	λ (-)
15	0.00	0.00	0.00	0.00	0.00
30	-3.75	7.00	8.85	0.25e-3	0.33
45	-2.25	11.00	7.24	0.32e-3	0.46
60	-1.75	15.00	5.36	0.81e-3	0.74
75	-0.50	14.50	3.39	0.62e-3	0.30

$$X_{\phi, \text{doy}} = \text{Avg}_{\phi} - \text{Amp}_{\phi} \cdot \cos\left((\text{doy} - 28) \frac{2\pi}{365.25}\right)$$

$$d_h^z = \frac{10^{-6} k_1 R}{g_m} \cdot P_0 \cdot \left(1 - \frac{\beta H}{T_0}\right)^{\frac{g}{R\beta}}$$

$$d_{nh}^z = \frac{10^{-6} (T_m k_2' + k_3) R}{g_m \lambda' - \beta R} \cdot \frac{e_0}{T_0} \cdot \left(1 - \frac{\beta H}{T_0}\right)^{\frac{\lambda' g}{R\beta} - 1}$$

(Leandro et al., 2006)

UNB3 uses mapping functions developed by A. E. Niell (1996)

UNB3m_pack matlab package available

[DRATE HZD DHMFDEL WZD DWMFDEL]
= UNB3M(LATRAD, HEIGHTM, DAYOYEAR, ELEVRAD)



GPT-3/VMF-3 Model

- <https://vmf.geo.tuwien.ac.at/>
- matlab package available
- Empirical GPT-3 model (table on the right ->)
- VMF3 fits the mapping function coefficients with ray tracing
- VMF3 includes height correction
- VMF3 accepts input of forecast/operational meteorological data (e.g., ECMWF)
- 1°x 1° zenith delay data updated every 6 hours available from the link above

(Landskron & Böhm, 2018)

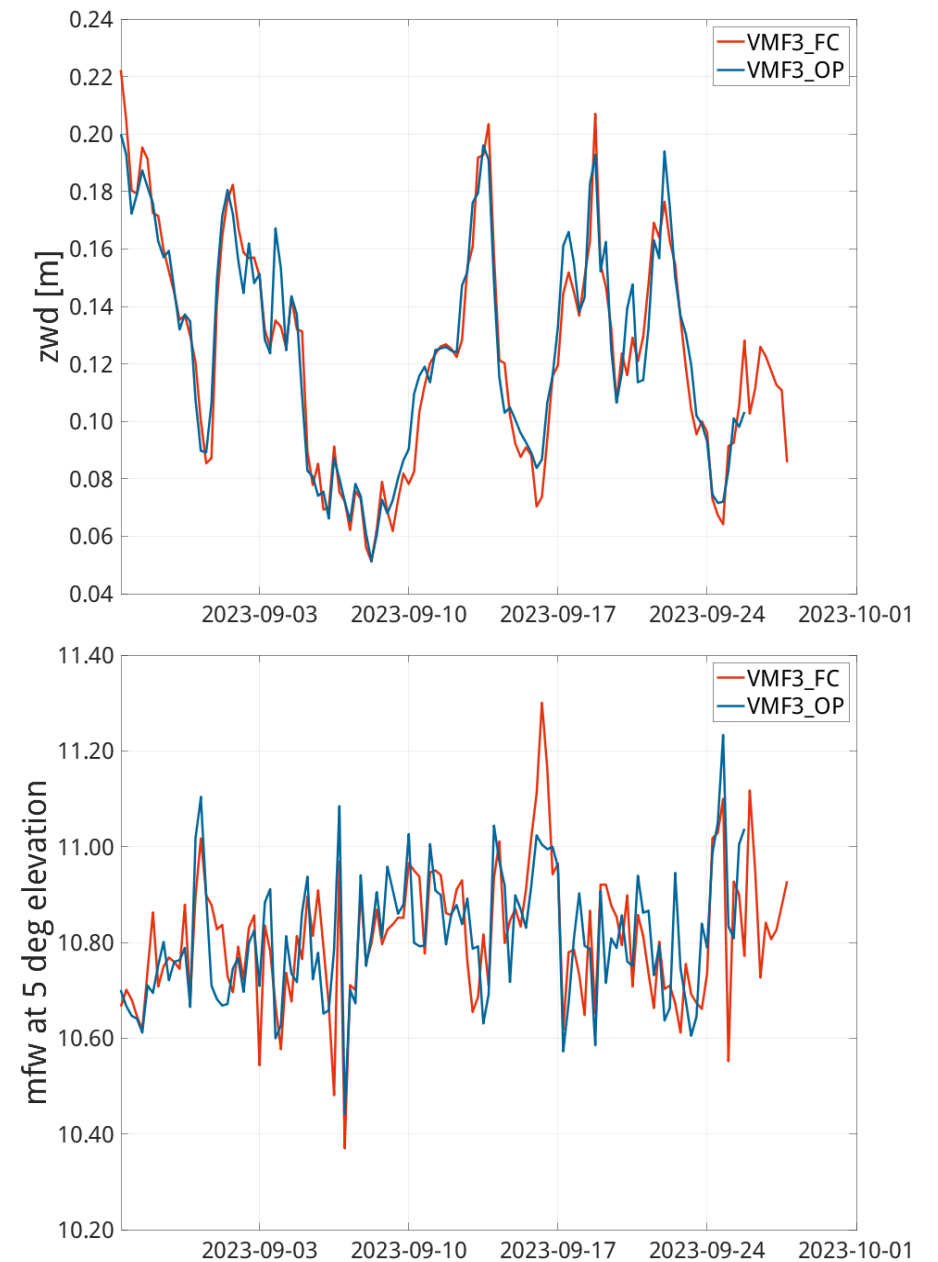
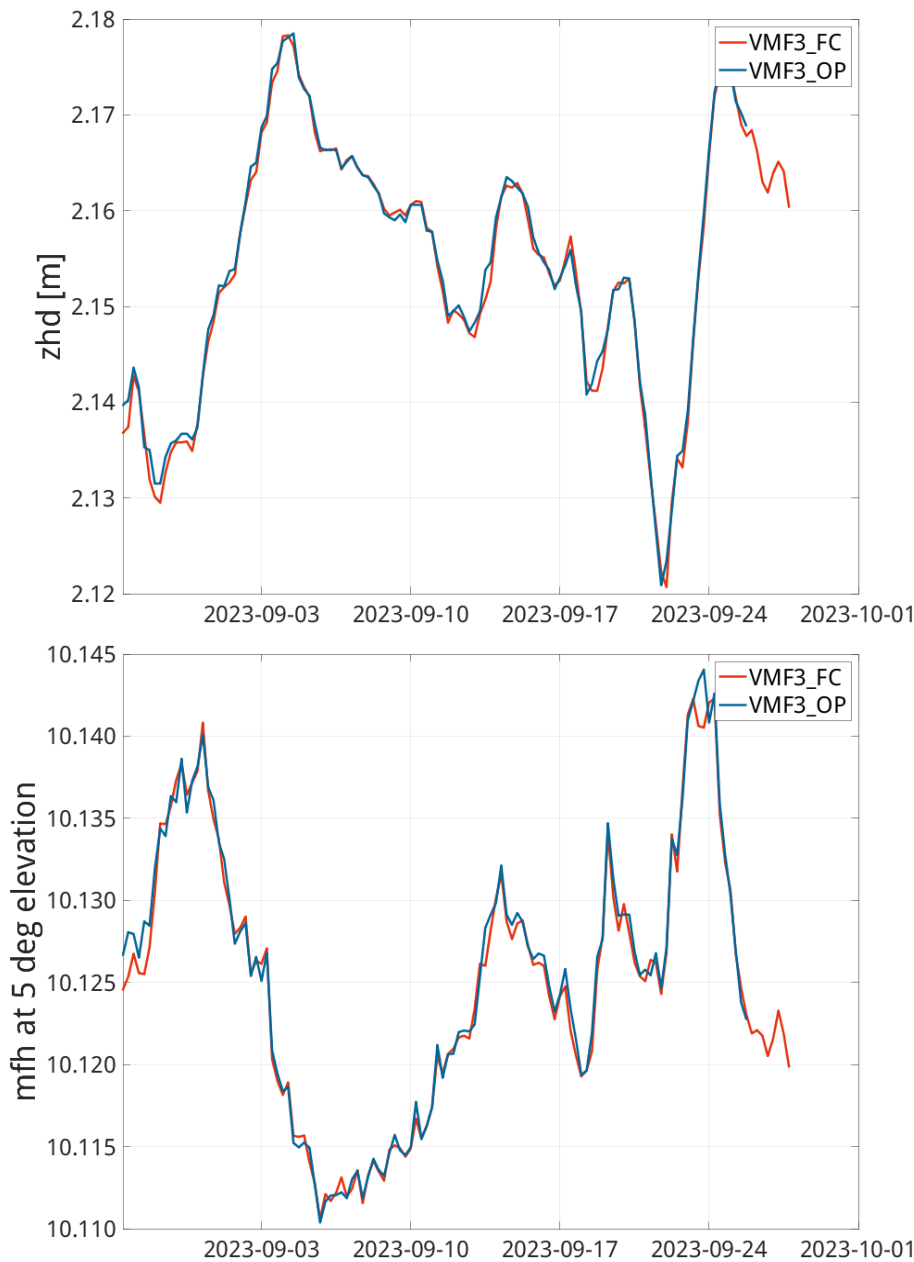
Table 8 A list of all input and output parameters of the empirical troposphere model GPT3

Symbol	Name	Unit
Input parameters		
mjd	Modified Julian date	–
φ	Geographic latitude	rad
λ	Geographic longitude	rad
h_{ell}	Ellipsoidal height	m
Output parameters		
p	Pressure	hPa
T	Temperature	°C
dT	Temperature lapse rate	K km ⁻¹
T_m	Mean temperature weighted with water vapor pressure	K
e	Water vapor pressure	hPa
a_h	Hydrostatic mapping function coefficient (valid at sea level)	–
a_w	Wet mapping function coefficient	–
λ	Water vapor decrease factor	–
N	Geoid undulation	m
G_{nh}	Hydrostatic north gradient	m
G_{eh}	Hydrostatic east gradient	m
G_{nw}	Wet north gradient	m
G_{ew}	Wet east gradient	m

Unless otherwise stated, all output quantities are valid for the ellipsoidal height h_{ell} specified in the input

GPT-3/VMF-3

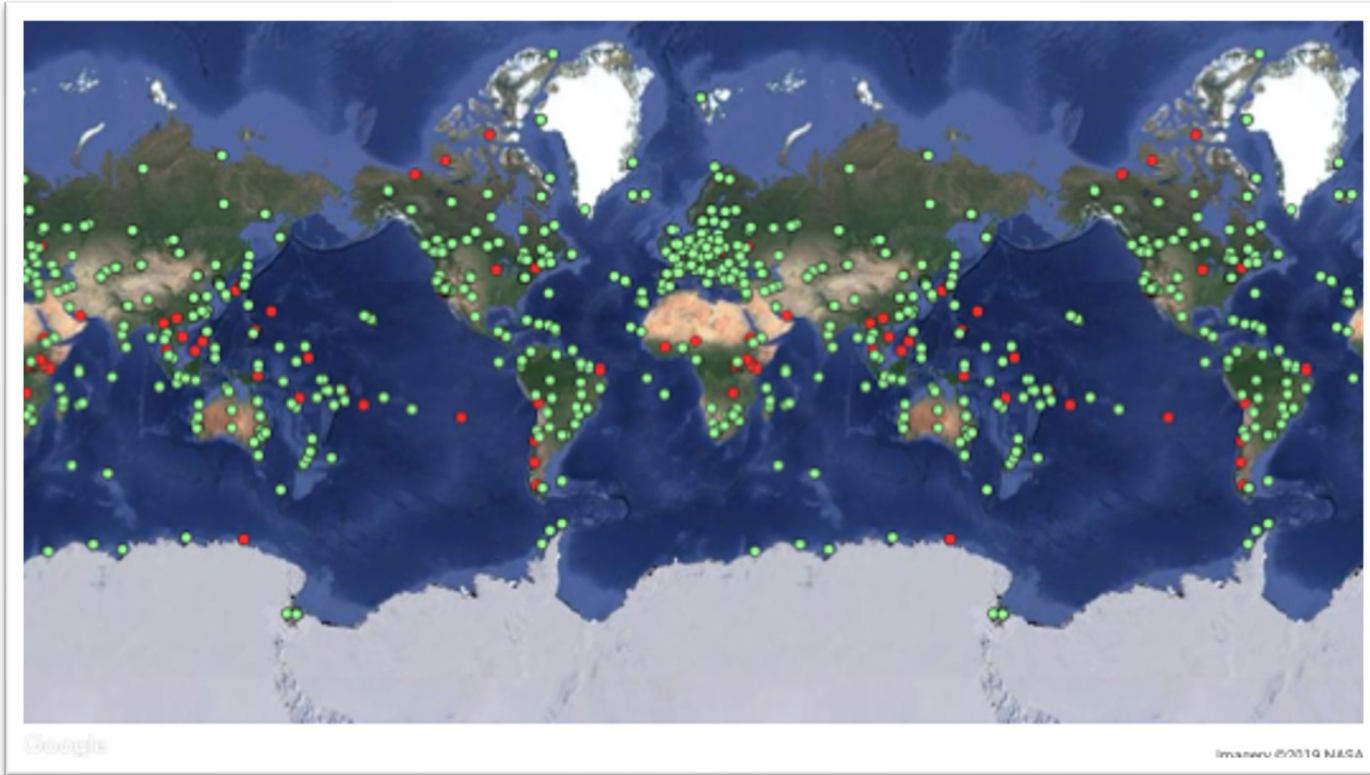
VMF3 parameters derived from operational Numerical Weather Models are depicted in blue (VMF3_OP), those derived from forecast Numerical Weather Models are depicted in red (VMF3_FC).



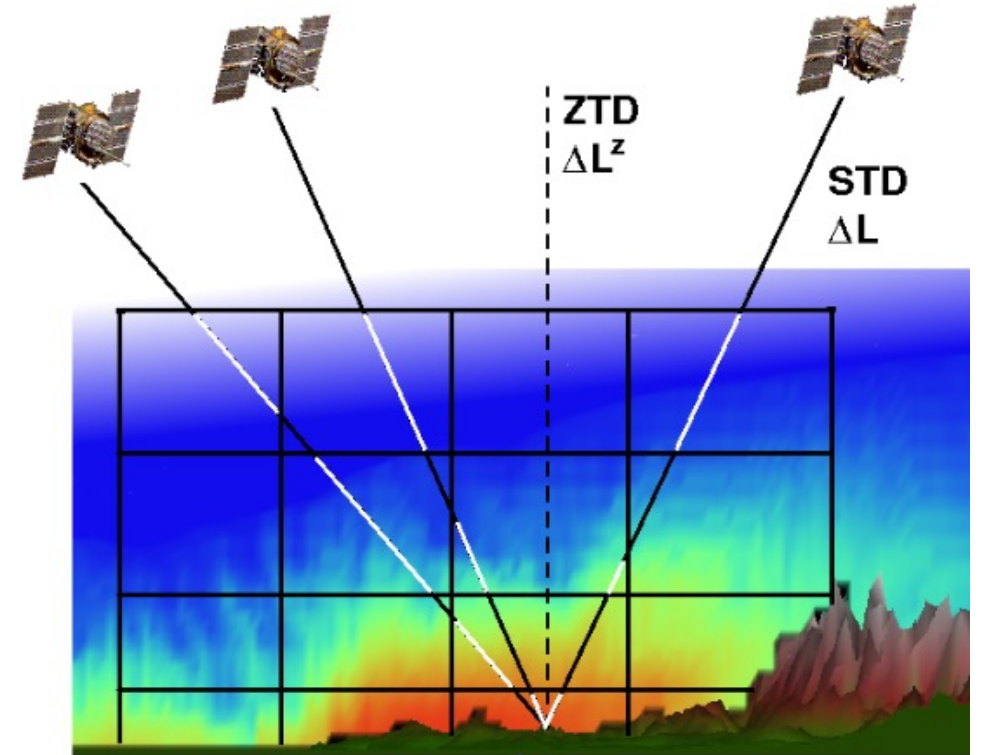
GNSS contributions to tropospheric mapping

Ground Networks

IGS Troposphere Products



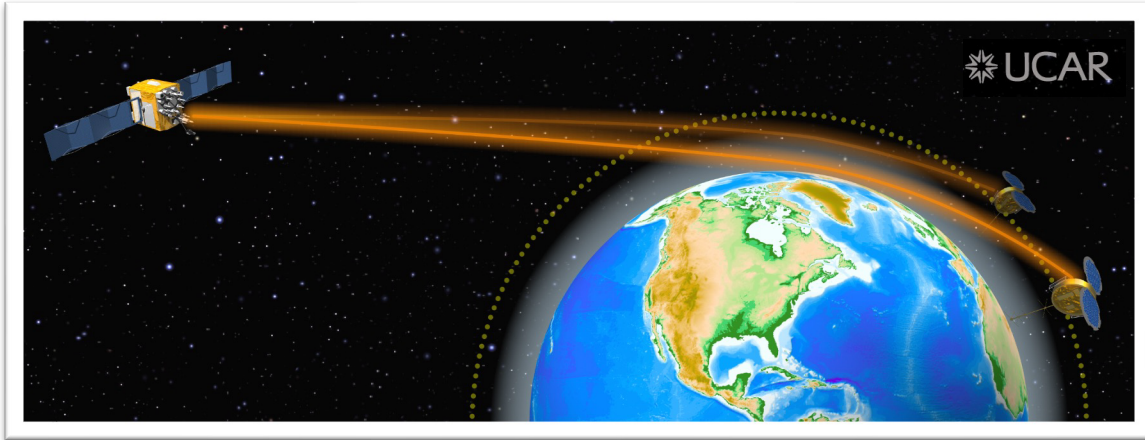
GNSS Troposphere Tomography



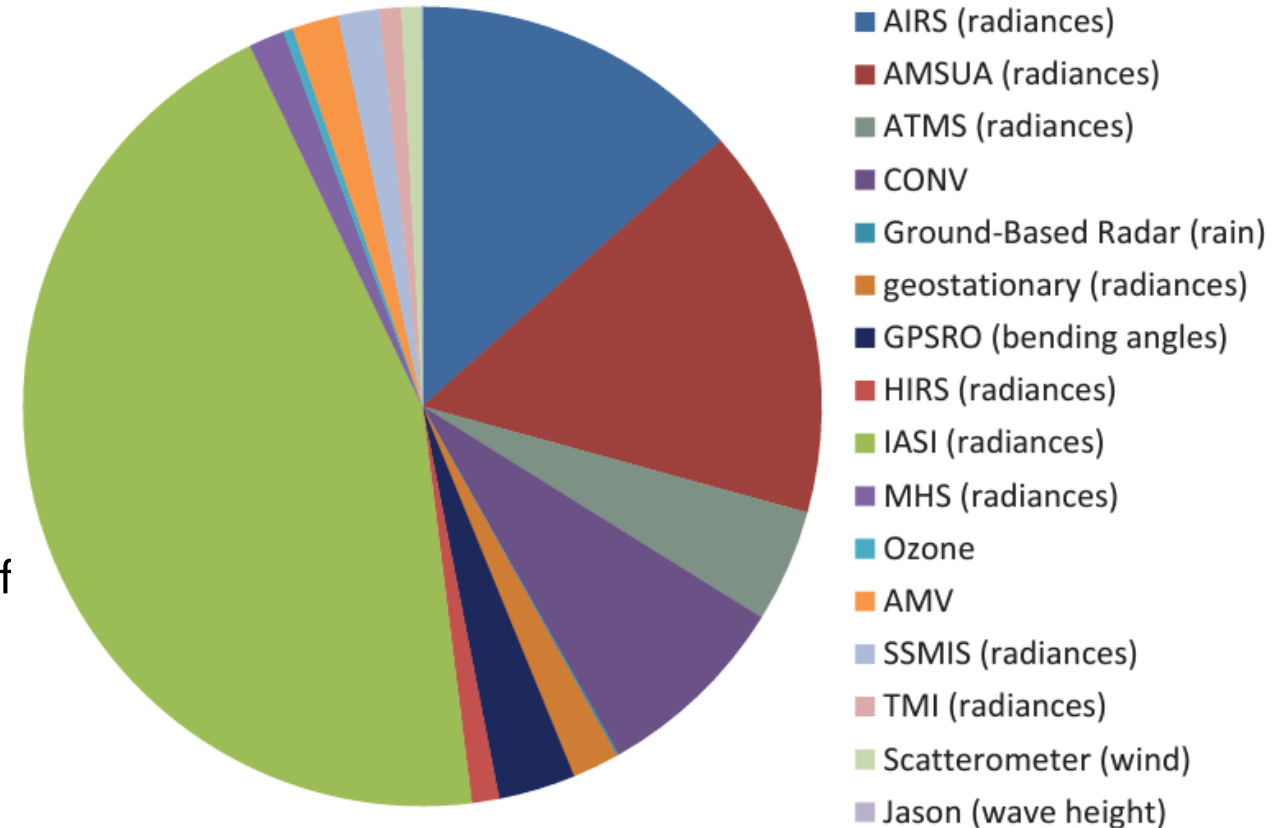
(Image Credit: GFZ)

GNSS contributions to tropospheric mapping

GNSS Radio Occultation

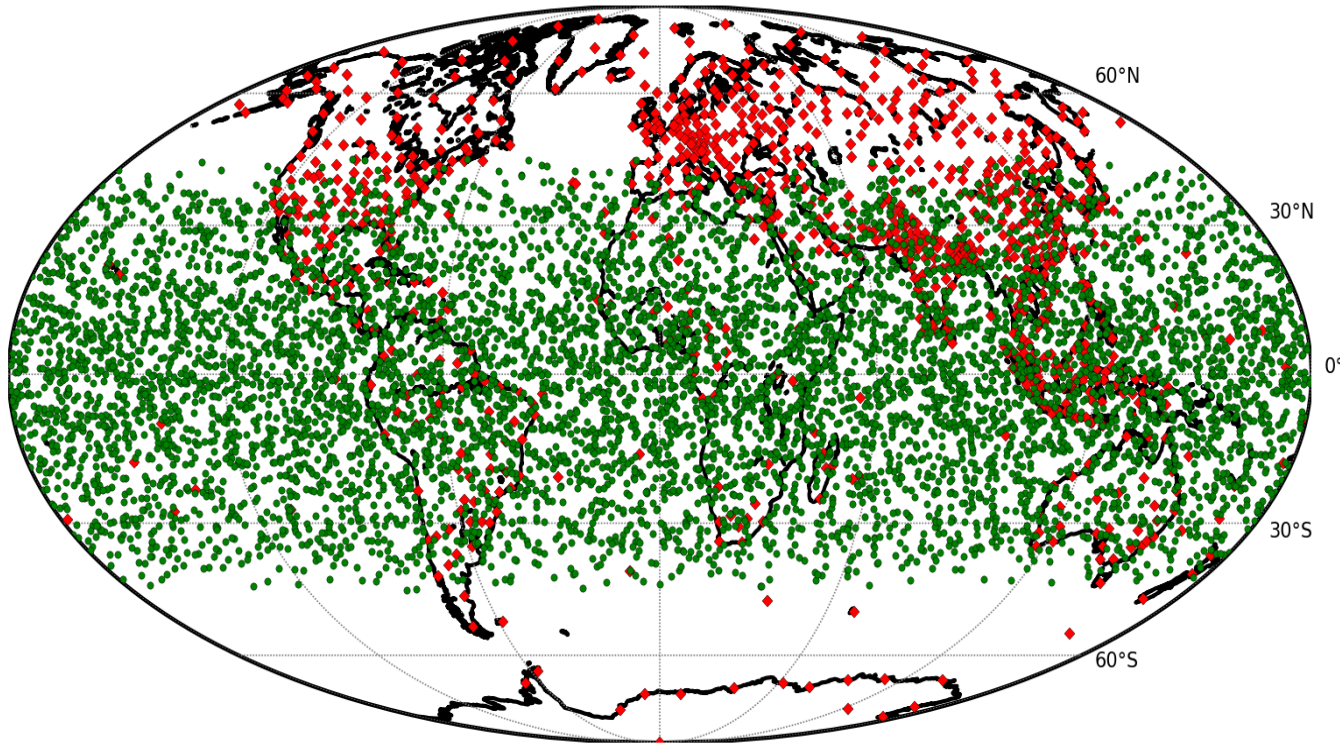


- GNSS-RO occultation measurements produce profiles of bending angle, refractivity, temp , humidity
- GNSS-RO contributes to numerical weather prediction (ECMWF) w/top 5 impact in reducing forecast error (Healy et al., **2014**)



GNSS contributions to tropospheric mapping

GNSS Radio Occultation



Expected daily GNSS-RO coverage from COSMIC-2 constellation shown in green. Radiosonde sites in red.

- FORMOSAT 3/COSMIC-1 (6) [2006-2020, retired], 1500-2500 profiles/day
- COSMIC-2 (6), launched 2019, ~5000 profiles/day
- GPS/MET, C/NOFS, CHAMP, SAC-C, GRACE, Metop, TerraSAR-X, etc.
- Commercial – Spire (30+), PlanetIQ, Geoptics

References

- **Misra & Enge Textbook Ch 5**
- Position, Navigation, and Timing Technologies in the 21st Century, Ed by Y.T.J. Morton, et al., Ch 30 GNSS for Neutral Atmosphere and Severe Weather Monitoring, Hugues Brenot, 2021.
- 2017 Springer Handbook of Global Navigation Satellite Systems, Springer Handbooks, Ch. 6 Atmospheric Signal Propagation – T. Hobiger and N. Jakowski
- UNB3 Model: <https://gge.ext.unb.ca/Resources/unb3m/unb3m.html>
- GPT3/VMF3 Model: <https://vmf.geo.tuwien.ac.at/>

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